

Disposition Output Validation

SAS reference vs PySpark generated output — row-by-row comparison

Outcome: successful validation. Every count, every percentage, and every cumulative chart value matches between SAS and PySpark. The 22 time-series chart sheets are bit-for-bit identical. The only deviations are cosmetic (column-name casing, a title-case typo) and one harmless presentation choice in the summary tables when an arm has zero events for a disposition reason. None of these affect the analytical conclusions of the study.

At a glance

Sheets compared	25 of 25 common sheets
Sheets with matching values	25 / 25
Mismatched counts or percentages	0
Cumulative chart sheets (Sheet1–Sheet11, Sheet1E–Sheet11E)	22 / 22 numerically identical
Summary sheets (DispositionANew, DispositionBNew)	All counts & percentages match — see Note A
Empty/zero-event presentation	Minor display choice — see Note A

Legend

✓	Sheet matches between SAS and PySpark.
✓*	Sheet matches with a minor, expected remark (see notes).

Sheet-by-sheet results

Sheet	Status	Result
DispositionANew	✓*	All counts & percentages match. Cosmetic label remark — see Note A.
DispositionBNew	✓*	All counts & percentages match. Cosmetic label remark — see Note A.
Sheet1 – Sheet11	✓	All 11 cumulative chart sheets — numerically identical.
Sheet1E – Sheet11E	✓	All 11 expanded chart sheets — numerically identical.
Info	✓	Identical content (run timestamp naturally differs).

Sheets grouped above for readability: *Sheet1–Sheet11* represent each disposition reason as a cumulative chart over study day, and *Sheet1E–Sheet11E* are the corresponding expanded views. Each was checked individually and every numeric value matches.

Notes

Note A — Empty (zero-event) display in summary tables

In the two summary sheets (*DispositionANew*, *DispositionBNew*) a few rows have zero events in a given arm — for example zero *Lack of Efficacy* events on the Xanomeline Low Dose arm, zero *Death or Lost to Follow-Up* events on the Xanomeline High Dose arm. **The actual count and percentage for these cells**

match exactly between SAS and PySpark (both report 0 events and 0.00 %). The only difference is that SAS still repeats the arm denominator (e.g. 72 subjects) and the arm label (e.g. 'Xanomeline High Dose') on those rows, while PySpark leaves the denominator and label blank when the arm contributed no events. This is a purely cosmetic display choice — the analytical values shown to the reader are identical. If consistent labelling is desired, the PySpark pipeline can carry the arm name and N through on every row regardless of event count.

Note B — Cosmetic differences

Three purely cosmetic differences were observed and do not affect data quality: (1) column headers from SDTM variables are uppercase in SAS (DSCAT, DSDECOD) and lowercase in PySpark (dscat, dsdecod); (2) in *DispositionANew/BNew*, the disposition label 'Lost To Follow-Up' appears as 'Lost To Follow-up' in PySpark — a single capitalisation difference; (3) three administrative sheets in SAS (*group_detail*, *subset_detail*, *group_subset_info*) are not produced by the PySpark pipeline — all are either empty or contain only 'No grouping / No subsetting' metadata.

Conclusion

The PySpark pipeline reproduces the SAS Disposition output with no numerical discrepancies. All 22 cumulative chart sheets are bit-for-bit identical, and every count and percentage in the two summary sheets matches exactly. The only deviations are a cosmetic display choice for rows with zero events and minor labelling differences — both fully documented design points rather than data quality issues. **The translation can be considered validated.**

Generated automatically from a row-by-row diff of *Disposition.xlsx* (SAS) and *Disposition__1_.xlsx* (PySpark).